

Are seed oils really bad for you?

REFERENCES	STYLE	TOKENS	VERIFICATION	COMPILED
70 verified	Scientific	≈15.2k	Crossref	28 Aug 2026

ABSTRACT

Ordinary culinary use of seed oils is unlikely to cause chronic disease and can improve cardiovascular risk when these oils replace fats richer in saturated or trans fatty acids. Controlled feeding studies consistently lower low-density-lipoprotein-related markers, and a pooled analysis of eight substitution trials reported a coronary-event risk ratio of 0.81 (95% confidence interval 0.70–0.95). Two recovered trials found no benefit or possible harm, but their old, high-dose, multifactorial interventions do not establish a class-wide hazard; prospective cohorts and human inflammation studies also fail to show broad harm. The conclusion applies to specified unsaturated oils used in ordinary cooking; it does not make ultra-processed foods healthy, prove that every oil is interchangeable, or support repeated heating until oil degrades.

01 Introduction

“Seed oil” is a culinary label, not a single intervention. Conventional soybean, corn, sunflower, safflower, canola, and grapeseed oils differ in fatty-acid composition, while high-oleic cultivars can resemble other monounsaturated-rich oils more than their conventional namesakes.^{1,2,3} The same oil can also appear in a controlled diet, a home-cooked meal, or a repeatedly used fryer. Those exposures share an ingredient name but not necessarily a biological effect.

The health question therefore has three parts. First, what nutrient did the oil replace? Second, did the observed outcome measure a causal risk factor, a clinical event, or only a plausible mechanism? Third, did processing and heat create a materially different exposure? Controlled substitutions, prospective cohorts, biomarkers, and heating experiments answer different parts of that sequence. The evidence agrees once those parts stay separate.

02 “Seed oils” is not a biologically uniform exposure

Oil composition determines which comparison is scientifically meaningful. Controlled diets test exchanges: unsaturated fat for saturated fat, one unsaturated oil for another, or fat for carbohydrate. They do not test an abstract category against an empty plate. Randomized substitutions and prospective models repeatedly show that the replacement nutrient changes both the direction and size of the estimate.^{4,5,6,7} Simply adding oil and calories licenses no equivalent inference.

Comparisons among unsaturated oils are usually smaller than comparisons with saturated fats. Trials of soybean,

canola, corn, high-oleic formulations, olive oil, and camellia oil often separate on lipid outcomes while remaining similar on endothelial or metabolic measures.^{8,9,10} That pattern argues against both blanket toxicity and a universal ranking. The specified oil, dose, and comparator matter.

High-oleic formulations make the category problem especially visible. Plant breeding can reduce an oil's linoleic-acid share while increasing oleic acid without changing the crop name on the front of the bottle.^{1,2} Conversely, a linoleic-rich oil used to displace butter answers a different question from the same oil added to an energy-surplus snack. A study can identify the first contrast while saying almost nothing about the second. Category-level claims lose that information before any outcome is measured.

Study design also changes what “exposure” means. Controlled feeding fixes energy and the replacement nutrient, improving attribution but narrowing external validity; cohorts observe habitual intake embedded in meals, increasing relevance while reintroducing confounding and measurement error.^{8,11,5} Agreement across those designs matters because their biases point in different directions. It supports a substitution conclusion but cannot convert an ingredient label into a dose-response curve. Neither stream licenses claims about oil consumed with excess energy or degraded through repeated heating.

The whole answer is organized around those exposure distinctions in [Figure 1](#). Its central point is practical: the ingredient name alone does not identify the health contrast being tested.

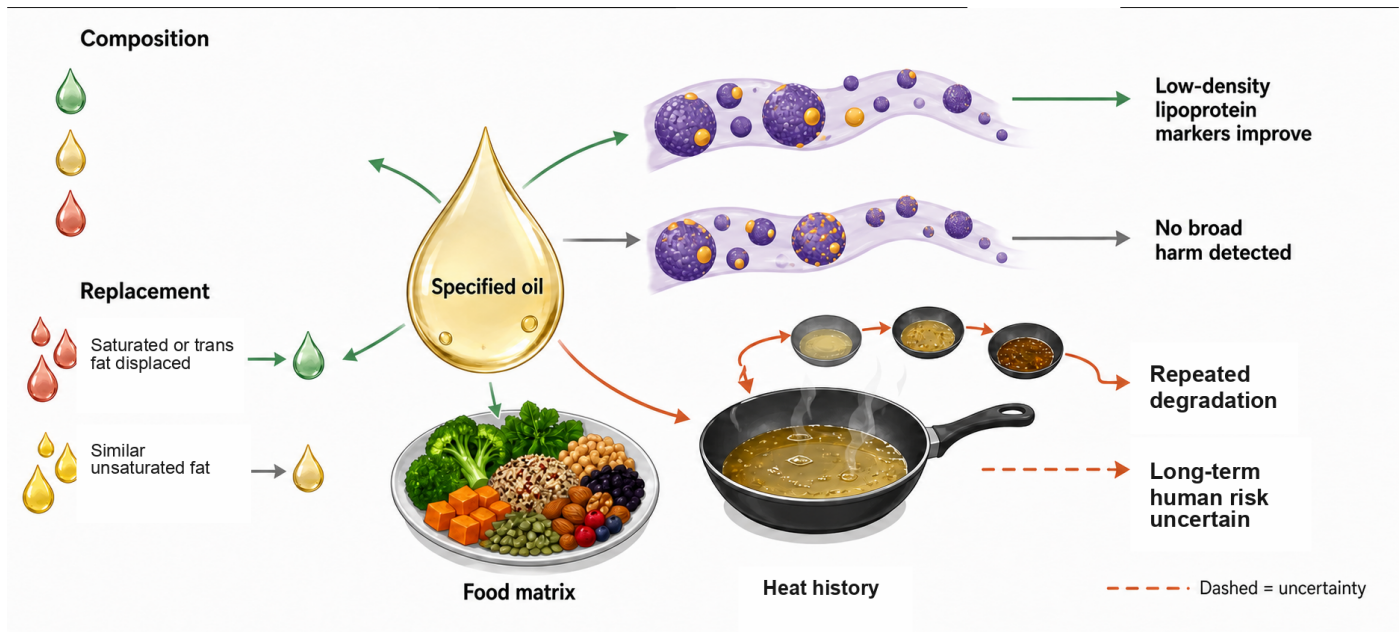


Figure 1. Ordinary use and thermal abuse are different exposures. The visual separates oil composition, the displaced nutrient, food context, and heating history before weighing human outcomes. Solid paths mark established comparisons; the muted dashed path marks uncertainty when oil is repeatedly degraded.^{1,5,12,13}

03 Replacing saturated fat with unsaturated oils lowers LDL-related risk markers

Controlled feeding studies provide the clearest causal result. When linoleic-acid-rich or other unsaturated oils replace saturated fats, low-density lipoprotein cholesterol and related atherogenic measures generally fall.^{14,11,15,16,17} Palm oil, which is botanically a fruit oil and relatively saturated, produces higher low-density-lipoprotein cholesterol than lower-saturated vegetable oils in pooled controlled comparisons.¹⁸ These trials identify a direction even when their absolute changes differ.

The causal chain is strongest at the lipid step. In a walnut and vegetable-oil crossover analysis, low-density-lipoprotein cholesterol was 6.9 mg/dL lower than with an oleic-acid comparison diet.¹⁹ Yet lipid improvement did not guar-

antee movement in every surrogate: a canola substudy reported no change in flow-mediated dilation ($P=0.81$).⁹ One marker can improve while another remains unchanged.

That distinction is not a contradiction. Lipoprotein concentration, vascular reactivity, blood pressure, and glucose regulation arise through overlapping but non-identical pathways, so a short feeding trial may move one before another. The controlled diets were also designed to isolate replacement effects, not to reproduce every food pattern in which an oil appears. Their strength is causal specificity. Their limitation is the distance from a surrogate measured over weeks to disease accumulated over decades.

The controlled comparisons below preserve the distinction between a robust lipid response and less consistent secondary endpoints.

COMPARISON	MAIN RESULT	BOUNDARY	SOURCE
Unsaturated vs saturated	LDL lower	Marker outcome	¹⁴
Vegetable oil vs oleic	LDL -6.9 mg/dL	Short crossover	¹⁹
Canola comparison	FMD $P=0.81$	Endothelial null	⁹
Palm vs lower-SFA oils	LDL higher	Palm only	¹⁸

Lipid trials therefore support substitution, but they cannot by themselves tell us how many myocardial infarctions an oil prevents. Clinical outcomes require a separate evidential step.

04 Randomized clinical outcomes exclude large harm but leave modest benefit uncertain

Randomized outcome syntheses do not show a large cardiovascular hazard from increasing polyunsaturated fat. Cochrane reviews of omega-6 or total polyunsaturated-fat in-

terventions found no consistent increase in cardiovascular events or mortality; some estimates favored fewer events, but trial quality, intervention mixtures, and precision limited confidence.^{12,20,21} A separate meta-analysis of eight trials, 13,614 participants, and 1,042 coronary events estimated a risk ratio of 0.81 (95% confidence interval 0.70–0.95) for polyunsaturated-fat substitution, or about 10% lower risk for each 5% of energy exchanged.²² Broader syntheses are more cautious, but none establishes broad toxicity.²³

The estimates in Figure 2 explain why reasonable reviewers differ about benefit yet agree that a large class-wide hazard is implausible. The pooled coronary-event estimate favors substitution, whereas two recovered old trials sit nearer to

or above the null for mortality. Their intervals and populations matter as much as their point estimates.

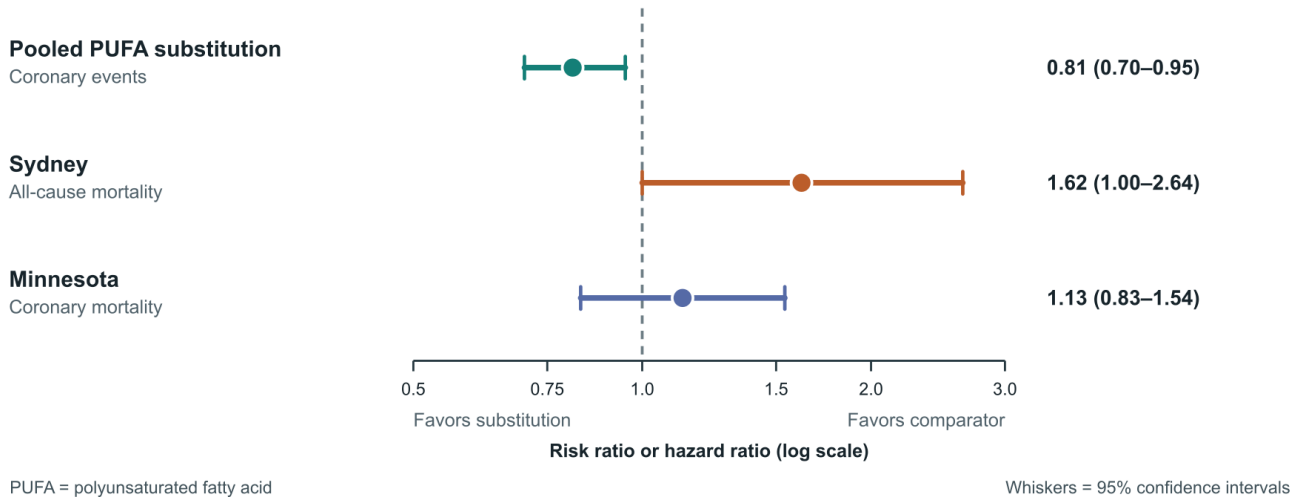


Figure 2. Clinical-event estimates disagree more than lipid outcomes. Exact log-scale positions show the pooled coronary-event benefit beside mortality estimates from the Sydney and Minnesota recovered trials. Different outcomes and trial contexts prevent a mechanical pooled verdict.^{22,24,25,12,20}

The clinical record thus supports at most a modest benefit from replacing saturated fat. Its more secure conclusion is narrower: randomized evidence does not reveal the large cardiovascular penalty claimed for ordinary seed-oil use.

Trial aggregation cannot remove differences in the interventions themselves. Older studies sometimes changed several fats at once, used margarines or products unlike present formulations, or failed to document background trans-fat exposure; follow-up and adherence also differed sharply.^{12,20,24,25} A pooled estimate therefore answers whether a family of substitutions tended to help, not whether every named oil produces an identical event reduction. The interval around the pooled result describes sampling uncertainty; it does not capture all of that clinical heterogeneity.

05 Two recovered trials do not generalize cleanly

The Sydney Diet Heart Study deserves attention because its recovered records point in the adverse direction. Among 458 men with established coronary disease, a safflower-oil intervention near 15% of energy was associated with all-cause mortality of 17.6% versus 11.8% (hazard ratio 1.62, 95% confidence interval 1.00–2.64); cardiovascular and coronary mortality estimates were similar.²⁴ The intervention also changed more than one dietary feature, records were incomplete, and the population already had coronary disease. This is a serious outlier with a narrow target population.

The Minnesota Coronary Experiment tells a different cautionary story. It randomized 9,423 institutionalized participants, but only 2,355 remained exposed for at least one

year. Serum cholesterol fell 13.8% in the intervention group versus 1.0% in controls without a mortality benefit; the updated coronary-mortality estimate was 1.13 (95% confidence interval 0.83–1.54).²⁵ Short exposure, attrition, missing records, institutional diets, and an age interaction make the result difficult to transport to long-term modern use.

Neither recovered trial can be ignored, and neither supplies a general estimate for fresh culinary oil. They weaken confidence in event reduction while leaving the much better replicated lipid effect intact.²⁶ That is why clinical benefit remains more uncertain than risk-factor improvement.

Trial age matters because the interventions, populations, and care contexts recorded in these studies differ from those in current routine use.^{24,25} Institutional residence or established coronary disease shaped baseline risk, and the interventions used products and background diets specific to their eras. Those differences do not invalidate randomization within each trial. They do restrict transport from the randomized contrast to present-day populations and products. The adverse estimates should therefore remain visible as uncertainty, not be promoted into a universal toxicity coefficient.

06 Prospective cohorts and biomarkers weigh against broad cardiovascular harm

Large prospective cohorts usually associate higher linoleic-acid intake with neutral or lower coronary and mortality risk. Meta-analyses of dietary intake are directionally consistent across diverse cohorts, although food-frequency measurement, residual confounding, and modeled substitu-

tions remain important limitations.^{27,28,29} Some cohorts are null, especially when the substitution is poorly specified.^{30,31} Observational evidence cannot prove benefit, but it does not resemble a strong common toxin signal.

Biomarkers reduce reliance on dietary recall and point the same way. Pooled analyses of circulating or tissue linoleic acid generally find neutral-to-inverse associations with cardiovascular events and all-cause mortality.^{32,33,34,35} A Framingham Offspring analysis followed 2,500 adults for up to 9.5 years, recording 245 cardiovascular events and 350 deaths; more recent blood-based studies extend the pattern without converting it into a randomized effect.^{36,37} Metabolism and genetics still influence every biomarker.

Biomarkers solve one measurement problem while creating another. They reduce misremembered intake and avoid asking participants to distinguish oils hidden in mixed dishes, but they also reflect absorption, endogenous metabolism, and the time window represented by the sampled tissue.^{32,33,35} Concordance across dietary and biomarker cohorts is therefore more informative than either stream alone. It lowers the likelihood that recall error created the entire association, while residual confounding and reverse causation remain possible.

Stroke findings make the convergence more concrete. Danish adipose linoleic acid in the highest versus lowest quartile was associated with ischemic-stroke hazard ratio 0.78 (95% confidence interval 0.65–0.93), while pooled UK circulating data reported odds ratio 0.82 (0.75–0.90) per standard deviation for stroke.^{38,39,40} Prospective substitution analyses also favor replacing saturated fat, though they do not extend automatically to hemorrhagic stroke.⁷ The cohort record strengthens the case against broad harm; it does not turn oil into a preventive drug.

07 Human trials do not show a consistent inflammatory response

The simplest inflammatory story predicts that more dietary linoleic acid yields more arachidonic acid and then more inflammatory mediators. Human metabolism is more regulated. Reviews of randomized trials find no consistent increase in C-reactive protein, interleukin-6, tumor-necrosis factor, or related conventional markers when linoleic acid rises.^{41,42,43} Human metabolic reviews also show that conversion to arachidonic acid is constrained and varies with genotype, background omega-3 intake, and disease state.^{44,45,3}

This does not erase biochemical plausibility. Linoleic and arachidonic acids feed specialized signaling and oxidation pathways, and some eicosanoids or oxylipins can promote inflammation in particular contexts.⁴⁴ Conventional blood markers may miss local tissue signaling. The inferential error is to treat a possible pathway as if it already measured net disease in people.

Dose and metabolic context further constrain the pathway. Fatty-acid enzymes process multiple substrates, and increasing one precursor does not force every downstream mediator to rise proportionally.^{44,45} A genotype or disease state may still shift that balance, which is why subgroup experiments remain useful. The current human trials answer a population-average question: they find no reproducible rise in the standard circulating markers at the tested intakes.^{41,42,43} They cannot declare every tissue pathway inactive.

Marker choice sets another boundary. Most eligible trials measured circulating markers over weeks or months, not every tissue-specific mediator or long-latency clinical event.^{41,42,43} A null average can coexist with transient local signaling, subgroup responses, or outcomes that were not sampled. The reverse inference is equally important: invoking an unmeasured mediator cannot overturn the measured nulls. Current trials directly argue against a large, consistent systemic response at the studied intakes, while leaving finer pathways and susceptible subgroups open. Longer controlled studies that pair repeated blood measurements with tissue or lipid-mediator profiling would narrow that residual uncertainty without mistaking mechanism for disease.

Clinical relevance remains a separate step. These marker trials were not designed or powered to detect long-latency disease, so their null results should not be recast as proof of zero clinical effect.^{41,42,43} They instead remove one proposed bridge in the broad-toxicity argument: a reproducible rise in conventional systemic inflammation. Linking any residual tissue signal to disease would require a study that measures the mediator, establishes its temporal response to a specified oil substitution, and then shows that the mediator predicts a patient-important outcome. No current experiment completes that chain.

Figure 3 separates the observed human marker results from the proposed mediator pathway. The dashed transition is deliberate: pathway chemistry warrants study, but it has not produced a reproducible systemic inflammatory response in controlled human feeding.

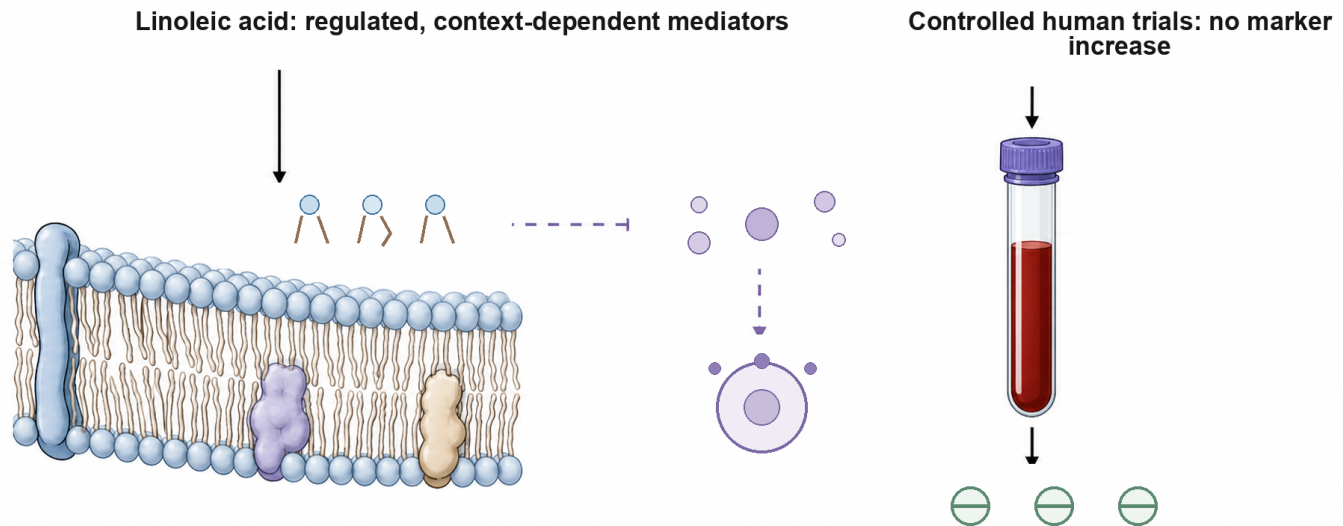


Figure 3. Plausible signaling does not equal systemic inflammation. Regulated conversion and mediator pathways remain biologically plausible, while controlled human studies do not show consistent rises in conventional inflammatory markers. Dashed arrows mark inferred or context-dependent steps.^{41,42,44,45}

Inflammation therefore illustrates a recurring pattern: mechanisms define what could happen, whereas controlled human measurements show what did happen under the tested conditions.

08 Metabolic outcomes are neutral to modestly favorable

Controlled human experiments do not show that plant-derived polyunsaturated fat worsens glycaemic control. A meta-analysis of 13 feeding trials estimated fasting glucose at -0.01 mmol/L (95% confidence interval -0.06 to 0.03), fasting insulin at -2.6 pmol/L (-4.9 to -0.2), and HOMA-IR at -0.12 (-0.23 to -0.01).⁴⁶ Omega-6-specific and diabetes-focused trials remain heterogeneous and often short.^{47,48} The measured effects are small.

Liver-fat experiments point in a compatible direction. Trials in abdominal obesity and overfeeding found that polyunsaturated fat produced less adverse liver-fat accumulation than saturated fat.^{49,50} A 12-month intervention in people with prediabetes or diabetes reduced liver fat by 1.46 percentage points (95% confidence interval -2.42 to -0.51) versus usual care.⁵¹ These selected populations do not support a

treatment claim for steatohepatitis, but they weigh against the idea that omega-6-rich oils inherently drive ectopic fat.

Prospective diabetes evidence is likewise neutral to favorable. Dose-response meta-analysis and biomarker-pattern studies associate higher linoleic-acid exposure with lower or unchanged type 2 diabetes risk.^{52,53} Among adults with diagnosed type 2 diabetes, replacing 2% of energy from saturated fat with linoleic acid was associated with cardiovascular-mortality hazard ratio 0.85 in two US cohorts.⁵⁴ Randomized prevention trials with hard endpoints are still missing.

The metabolic evidence also depends on what the intervention displaced. A diet that holds calories constant and exchanges saturated for polyunsaturated fat tests tissue handling of fatty-acid class; it does not test the consequences of adding fried food to an unchanged diet.^{49,50,46} The liver-fat experiments are particularly useful because they compare storage under controlled energy conditions, yet their sample sizes and selected populations limit generalization. Directional agreement across liver fat, fasting insulin, and prospective diabetes risk is reassuring; magnitude remains modest.

OUTCOME	ESTIMATE	BOUNDARY	SOURCE
Fasting glucose	-0.01 mmol/L	Short trials	⁴⁶
HOMA-IR	-0.12	Surrogate	⁴⁶
Liver fat	-1.46 points	Selected adults	⁵¹
Any cancer	RR 1.21	Wide interval	⁵⁵
Cancer biomarker	RR 0.92	Observational	⁵⁶

Across glucose, insulin resistance, and liver fat, the human data contradict a large adverse metabolic effect. The remaining uncertainty concerns long-term clinical outcomes

and individual susceptibility, not a consistent direction of harm.

09 Human evidence does not establish higher cancer risk

Cancer evidence is weaker than cardiovascular evidence because latency is long and randomized outcomes are sparse. A synthesis of 70 prospective reports found a neutral association for dietary omega-6 intake (relative risk 1.02, 95% confidence interval 0.99–1.05) and a modest inverse association for blood omega-6 biomarkers (0.92, 0.86–0.98).⁵⁶ Randomized evidence for any cancer remained imprecise: relative risk 1.21 (0.96–1.53) among 4,272 participants.⁵⁵ That interval permits benefit or harm.

Site-specific prospective meta-analyses do not show a consistent signal for breast or prostate cancer.^{57,58} They cannot exclude effects in molecular subtypes or at unusual exposures, and measurement error is substantial. The available human literature therefore supports neither a general carcinogenic label nor a confident protective claim.

Long latency makes one-time dietary questionnaires especially vulnerable to dilution. Tumors grouped under one site can also differ biologically, so a null total may conceal smaller opposing subtype effects.^{56,57,58} Repeated biomarkers and molecular tumor classification would improve the next generation of studies. Until such data exist, mechanistic arguments should not outrun the wide human intervals.

Cancer is the clearest place to preserve uncertainty. Neutral observational averages and underpowered randomized endpoints narrow the plausible range but do not close it.

10 Heat history creates a separate exposure

Polyunsaturated oils oxidize more readily than monounsaturated-rich oils during prolonged high heat. Laboratory and

food-chemistry studies detect aldehydes and other degradation products as temperature, duration, and reuse increase.^{59,60} Antioxidants, high-oleic formulations, replenishment, and temperature control materially change the rate.¹ Chemical susceptibility is real; its clinical meaning depends on exposure.

The temperature-response evidence describes a gradient. A heating meta-analysis found little trans-fat formation below 200 degrees C, while total trans fat rose by 0.38 percentage points per 10 degrees C between 200 and 240 degrees C (95% confidence interval 0.20–0.55).¹³ Repeated commercial or household frying adds storage, oxygen, moisture, and cumulative cycles. One study of restaurant fries found roughly 10–25 parts per million of each monitored aldehyde class, with large variation by oil and protocol.⁶¹

Human outcome evidence for degraded frying oil remains limited. Reviews, occupational exposure studies, and short postprandial experiments justify minimizing repeated reuse, but long-term randomized trials do not exist.^{62,63,64} Figure 4 marks the boundary without pretending that one sauté and an abused fryer are equivalent.

Heating is also a process rather than a single temperature. Oil volume, pan geometry, oxygen exposure, water released from food, cooling, storage, and replenishment all change degradation.^{59,13,61} Visible smoke is a practical warning that the oil has exceeded an appropriate cooking range, but a universal household threshold cannot be inferred from the available experiments. Reuse adds time at temperature and preserves reaction products between cooking cycles. That cumulative history is the exposure of concern.

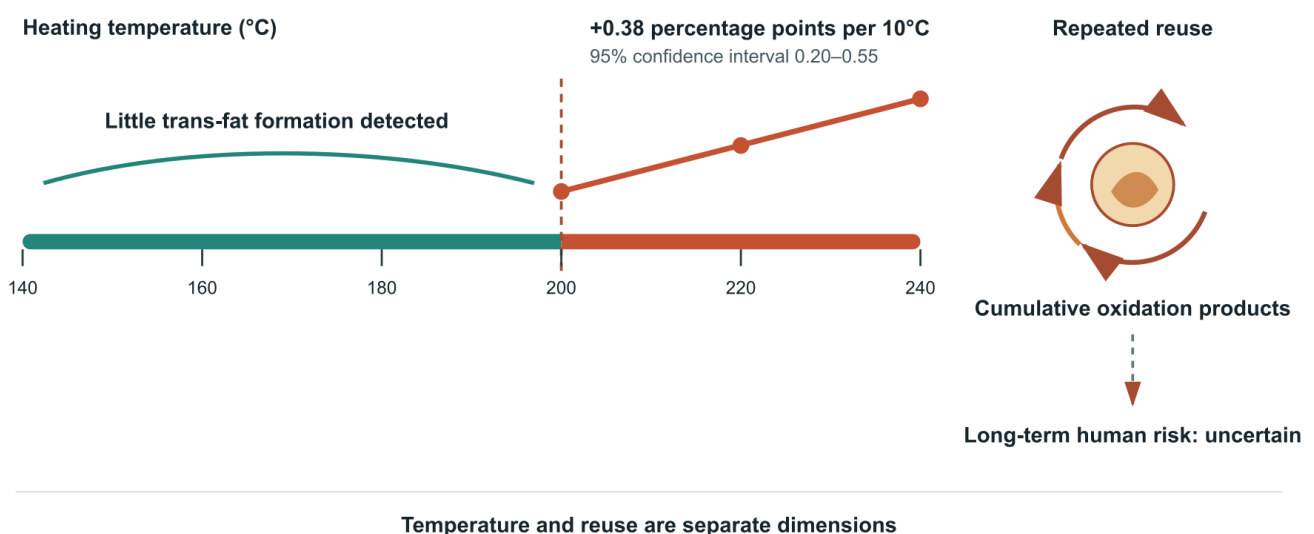


Figure 4. Heat and reuse change the exposure. The exact temperature response applies to total trans-fat formation in the heating meta-analysis; the separate reuse path represents cumulative oxidation rather than a quantified chronic-disease risk. Fresh, controlled cooking and repeated degradation remain distinct.^{13,59,61,62}

Heating chemistry therefore sets a sensible precautionary limit. It does not overturn the evidence on ordinary substitution with fresh oil.

11 The omega-6-to-omega-3 ratio can mislead

A ratio cannot reveal whether its numerator rose, its denominator fell, or both changed. In UK Biobank, the highest versus lowest plasma omega-6-to-omega-3 ratio quintile was associated with all-cause-mortality hazard ratio 1.26 (95% confidence interval 1.15–1.38), even though omega-6

and omega-3 concentrations were each independently inversely associated with mortality.⁶⁵ Shared metabolic pathways make the ratio biologically interesting, but they do not repair this mathematical ambiguity.⁴⁴

Figure 5 shows how the adverse ratio and favorable components can coexist. A high ratio may identify too little omega-3 rather than too much omega-6; the ratio alone cannot choose between those explanations.

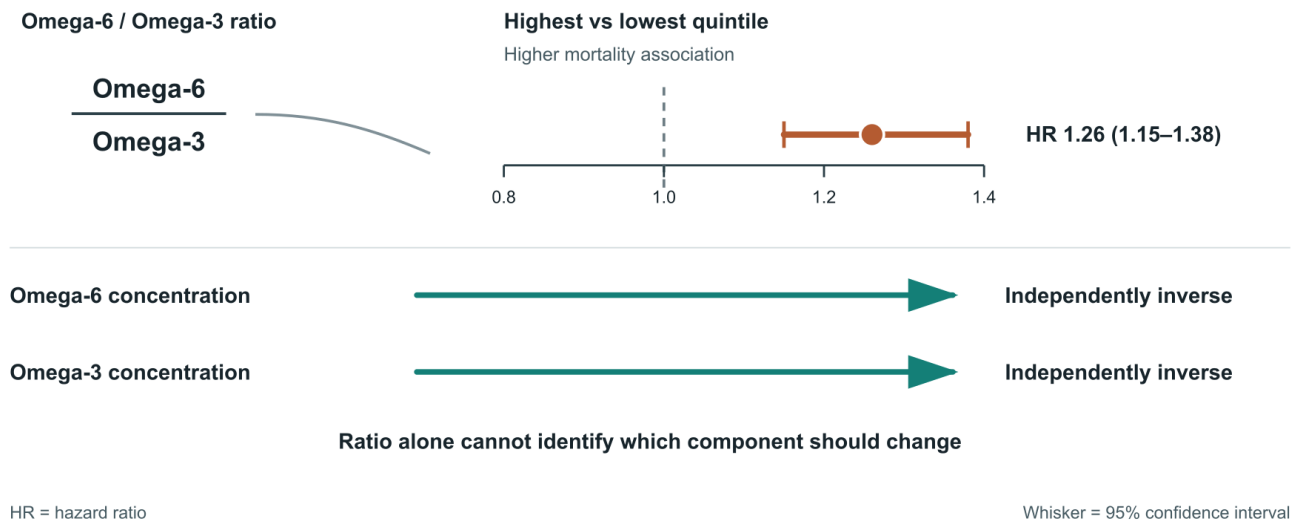


Figure 5. The ratio can reverse the component-level interpretation. The reported ratio association is adverse, while omega-6 and omega-3 concentrations are each inversely associated with mortality. The ratio does not identify which component should change.^{65,44}

This is a measurement problem with practical consequences. Evaluating both fatty-acid levels is more informative than treating the ratio as a diagnosis.

The same arithmetic applies across individuals. Two people can share a ratio while having very different concentrations of both fatty acids, and an intervention can improve the denominator without changing the numerator. Ratio-only advice hides those paths. Component measures preserve them.

12 Food matrix and processing alter what an oil label means

Foods containing similar fatty-acid replacements need not have identical effects. In a controlled crossover study, whole walnuts changed central diastolic pressure by −1.78 mm Hg compared with 0.15 mm Hg for an oleic-acid replacement ($P=0.04$), while systolic pressure and arterial stiffness were unchanged.⁶⁶ Broad food-substitution syntheses similarly suggest that the source food carries information beyond the isolated fatty acid.⁵ The matrix can matter without making every whole food superior on every endpoint.

The same distinction prevents seed-oil intake from serving as a reliable proxy for ultra-processed food. Seed oils occur in controlled diets that improve lipids and in minimally processed foods, as well as in products high in refined

starch, sodium, and energy density.^{8,11} No trial decomposes every feature of an ultra-processed pattern into an oil-specific share. Assigning the pattern's entire association to one ingredient exceeds the evidence.

New molecular studies strengthen the substitution story without settling clinical outcomes. In linked lipidomic analyses, 69 plasma lipid features predicted at least one outcome, and 17 of 19 diet-responsive risk-associated lipids moved favorably during unsaturated-fat substitution.^{67,68} These specialized markers connect controlled feeding with prospective risk patterns. Clinical events remain the test that matters most.

Processing likewise bundles exposures that trials can separate only partially. Grinding, emulsification, starch refinement, sodium, energy density, flavor engineering, and portion size can travel with an oil ingredient but need not do so.^{66,5} A label-based observational analysis that lacks those details cannot assign their combined association to the oil alone. Controlled feeding isolates the fatty-acid contrast more cleanly, while whole-food trials preserve more of the matrix. Both designs are needed because each removes a different ambiguity.

Taken together, the food evidence supports a simple direction rather than a universal product ranking. Replacing butter or other saturated animal fats with mostly unsatur-

ated plant sources aligns with substitution cohorts and lipid trials; replacing unsaturated oil with butter generally does not.^{5,14,18} A modeled butter-to-olive-oil substitution was associated with cardiovascular hazard ratio 0.96 (95% confidence interval 0.95–0.98), but food context and equal calories remain part of that estimate.⁵

13 Conclusion

The literature repeatedly separates at the same boundary: specified substitutions in people produce a different answer from mechanisms detached from dose, food, and heating history. Controlled diets lower atherogenic lipids; clinical-event trials allow modest benefit but retain uncertainty; cohorts, biomarkers, inflammation trials, metabolic experiments, and cancer syntheses do not reveal a coherent broad-harm signal. Repeated thermal degradation is the exception that deserves separate precaution.^{69,70}

For most people, the evidence favors mostly unsaturated oils when they replace saturated or trans fats, avoidance of repeated reuse, and judgment of the whole food rather than its ingredient list alone. The decisive remaining experiment would specify the oil, match calories and processing, sustain exposure for years, and measure clinical events. Until then, “seed oils” is too broad a category to carry the toxic verdict often assigned to it.

References

70 · VERIFIED VIA CROSSREF

1. Rosqvist F, Niinistö S (2024) Fats and oils – a scoping review for Nordic Nutrition Recommendations 2023. *Food & Nutrition Research*. <https://doi.org/10.29219/fnr.v68.10487>
2. Huth PJ, Fulgoni VL, Larson BT (2015) A Systematic Review of High-Oleic Vegetable Oil Substitutions for Other Fats and Oils on Cardiovascular Disease Risk Factors: Implications for Novel High-Oleic Soybean Oils. *Advances in Nutrition*. <https://doi.org/10.3945/an.115.008979>
3. Maki KC, Eren F, Cassens ME, Dicklin MR, Davidson MH (2018) ω-6 Polyunsaturated Fatty Acids and Cardiometabolic Health: Current Evidence, Controversies, and Research Gaps. *Advances in Nutrition*. <https://doi.org/10.1093/advances/nmy038>
4. Mozaffarian D, Clarke R (2009) Quantitative effects on cardiovascular risk factors and coronary heart disease risk of replacing partially hydrogenated vegetable oils with other fats and oils. *European Journal of Clinical Nutrition*. <https://doi.org/10.1038/sj.ejcn.1602976>
5. Neuenschwander M, Stadelmaier J, Eble J, Grummich K, Szczerba E, Kiesswetter E, Schlesinger S, Schwingshackl L (2023) Substitution of animal-based with plant-based foods on cardiometabolic health and all-cause mortality: a systematic review and meta-analysis of prospective studies. *BMC Medicine*. <https://doi.org/10.1186/s12916-023-03093-1>
6. Farvid MS, Ding M, Pan A, Sun Q, Chiuve SE, Steffen LM, Willett WC, Hu FB (2014) Dietary Linoleic Acid and Risk of Coronary Heart Disease: A Systematic Review and Meta-Analysis of Prospective Cohort Studies. *Circulation*. <https://doi.org/10.1161/circulationaha.114.010236>
7. Venø SK, Schmidt EB, Jakobsen MU, Lundbye-Christensen S, Bach FW, Overvad K (2017) Substitution of Linoleic Acid for Other Macronutrients and the Risk of Ischemic Stroke. *Stroke*. <https://doi.org/10.1161/strokeaha.117.017935>
8. Wu MY, Du MH, Wen H, Wang WQ, Tang J, Shen LR (2022) Effects of n-6 PUFA-rich soybean oil, MUFA-rich olive oil and camellia seed oil on weight and cardiometabolic profiles among Chinese women: a 3-month double-blind randomized controlled-feeding trial. *Food & Function*. <https://doi.org/10.1039/d1fo03759e>
9. Davis KM, Petersen KS, Bowen KJ, Jones PJH, Taylor CG, Zahradka P, Letourneau K, Perera D, Wilson A, Wagner PR, Kris-Etherton PM, West SG (2022) Effects of Diets Enriched with Conventional or High-Oleic Canola Oils on Vascular Endothelial Function: A Sub-Study of the Canola Oil Multi-Centre Intervention Trial 2 (COMIT-2), a Randomized Crossover Controlled Feeding Study. *Nutrients*. <https://doi.org/10.3390/nu14163404>
10. Pourrajab B, Sharifi-Zahabi E, Soltani S, Shahinfar H, Shidfar F (2023) Comparison of canola oil and olive oil consumption on the serum lipid profile in adults: a systematic review and meta-analysis of randomized controlled trials. *Critical Reviews in Food Science and Nutrition*. <https://doi.org/10.1080/10408398.2022.2100314>
11. Chen CG, Wang P, Zhang ZQ, Ye YB, Zhuo SY, Zhou Q, Chen YM, Su YX, Zhang B (2020) Effects of plant oils with different fatty acid composition on cardiovascular risk factors in moderately hypercholesteremic Chinese adults: a randomized, double-blinded, parallel-designed trial. *Food & Function*. <https://doi.org/10.1039/d0fo00875c>
12. Hooper L, Al-Khudairy L, Abdelhamid AS, Rees K, Brainard JS, Brown TJ, Ajabnoor SM, O'Brien AT, Winstanley LE, Donaldson DH, Song F, Deane KH (2018) Omega-6 fats for the primary and secondary prevention of cardiovascular disease. *Cochrane Database of Systematic Reviews*. <https://doi.org/10.1002/14651858.cd011094.pub4>
13. Bhat S, Maganja D, Huang L, Wu JHY, Marklund M (2022) Influence of Heating during Cooking on Trans Fatty Acid Content of Edible Oils: A Systematic Review and Meta-Analysis. *Nutrients*. <https://doi.org/10.3390/nu14071489>
14. Iggman D, Gustafsson IB, Berglund L, Vessby B, Marckmann P, Risérus U (2011) Replacing dairy fat with rapeseed oil causes rapid improvement of hyperlipidaemia: a randomized controlled study. *Journal of Internal Medicine*. <https://doi.org/10.1111/j.1365-2796.2011.02383.x>
15. Maki KC, Lawless AL, Kelley KM, Kaden VN, Geiger CJ, Dicklin MR (2015) Corn oil improves the plasma lipoprotein lipid profile compared with extra-virgin olive oil consumption in men and women with elevated cholesterol: Results from a randomized controlled feeding trial. *Journal of Clinical Lipidology*. <https://doi.org/10.1016/j.jacl.2014.10.006>
16. Vafeiadou K, Weech M, Altowajiri H, Todd S, Yaqoob P, Jackson KG, Lovegrove JA (2015) Replacement of saturated with unsaturated fats had no impact on vascular function but beneficial effects on lipid biomarkers, E-selectin, and blood pressure: results from the randomized, controlled Dietary Intervention and VAScular function (DIVAS) study. *The American Journal of Clinical Nutrition*. <https://doi.org/10.3945/ajcn.114.097089>
17. Binkoski AE, Kris-Etherton PM, Wilson TA, Mountain ML, Nicolosi RJ (2005) Balance of Unsaturated Fatty Acids Is Important to a Cholesterol-Lowering Diet: Comparison of Mid-Oleic Sunflower Oil and Olive Oil on Cardiovascular Disease Risk Factors. *Journal of the American Dietetic Association*. <https://doi.org/10.1016/j.jada.2005.04.009>
18. Sun Y, Neelakantan N, Wu Y, Lote-Oke R, Pan A, van Dam RM (2015) Palm Oil Consumption Increases LDL Cholesterol Compared with Vegetable Oils Low in Saturated Fat in a Meta-Analysis of Clinical Trials. *The Journal of Nutrition*. <https://doi.org/10.3945/jn.115.210575>
19. Tindall AM, Kris-Etherton PM, Petersen KS (2020) Replacing Saturated Fats with Unsaturated Fats from Walnuts or Vegetable Oils Lowers Atherogenic Lipoprotein Classes Without Increasing Lipoprotein(a). *The Journal of Nutrition*. <https://doi.org/10.1093/jn/nxz313>
20. Abdelhamid AS, Martin N, Bridges C, Brainard JS, Wang X, Brown TJ, Hanson S, Jimoh OF, Ajabnoor SM, Deane KH, Song F, Hooper L (2018) Polyunsaturated fatty acids for the primary and secondary prevention of cardiovascular disease. *Cochrane Database of Systematic Reviews*. <https://doi.org/10.1002/14651858.cd012345.pub2>
21. Al-Khudairy L, Hartley L, Clar C, Flowers N, Hooper L, Rees K (2015) Omega 6 fatty acids for the primary prevention of cardiovascular disease. *Cochrane Database of Systematic Reviews*. <https://doi.org/10.1002/14651858.cd011094.pub2>
22. Mozaffarian D, Micha R, Wallace S (2010) Effects on Coronary Heart Disease of Increasing Polyunsaturated Fat in Place of Saturated Fat: A Systematic Review and Meta-Analysis of Randomized Controlled Trials. *PLoS Medicine*. <https://doi.org/10.1371/journal.pmed.1000252>
23. Mazidi M, Shekoohi N, Katsiki N, Banach M, Meta-analysis Collaboration (LBPMC) Group TLBP (2021) Omega-6 fatty acids and the Risk of Cardiovascular Disease: Insights from a Systematic Review and Meta-Analysis of Randomized Controlled Trials and a Mendelian Randomization Study. *Archives of Medical Science*. <https://doi.org/10.5114/aoms/136070>
24. Ramsden CE, Zamora D, Leelarthaepin B, Majchrzak-Hong SF, Faurot KR, Suchindran CM, Ringel A, Davis JM, Hibbeln JR (2013) Use of dietary linoleic acid for secondary prevention of coronary heart disease and death: evaluation of recovered data from the Sydney Diet Heart Study and updated meta-analysis. *BMJ*. <https://doi.org/10.1136/bmj.e8707>
25. Ramsden CE, Zamora D, Majchrzak-Hong S, Faurot KR, Broste SK, Frantz RP, Davis JM, Ringel A, Suchindran CM, Hibbeln JR (2016) Re-evaluation of the traditional diet-heart hypothesis: analysis of recovered data from Minnesota Coronary Experiment (1968-73). *BMJ*. <https://doi.org/10.1136/bmj.i1246>
26. Schwab U, Lauritzen L, Tholstrup T, Haldorsson TI, Riserus U, Uusitupa M, Becker W (2014) Effect of the amount and type of dietary fat on cardiometabolic risk factors and risk of developing type 2 diabetes, cardiovascular diseases, and cancer: a systematic review. *Food & Nutrition Research*. <https://doi.org/10.3402/fnr.v58.25145>

27. Li J, Guasch-Ferré M, Li Y, Hu FB (2020) Dietary intake and biomarkers of linoleic acid and mortality: systematic review and meta-analysis of prospective cohort studies. *The American Journal of Clinical Nutrition*. <https://doi.org/10.1093/ajcn/nqz349>
28. Wang DD (2018) Dietary n-6 polyunsaturated fatty acids and cardiovascular disease: Epidemiologic evidence. *Prostaglandins, Leukotrienes and Essential Fatty Acids*. <https://doi.org/10.1016/j.plefa.2018.05.003>
29. Sadeghi R, Norouzzadeh M, HasanRashedi M, Jamshidi S, Ahmadi H, Alemrajabi M, Vafa M, Teymoori F (2025) Dietary and circulating omega-6 fatty acids and their impact on cardiovascular disease, cancer risk, and mortality: a global meta-analysis of 150 cohorts and meta-regression. *Journal of Translational Medicine*. <https://doi.org/10.1186/s12967-025-06336-2>
30. de Goede J, Geleijnse JM, Boer JMA, Kromhout D, Verschuren WMM (2012) Linoleic acid intake, plasma cholesterol and 10-year incidence of CHD in 20 000 middle-aged men and women in the Netherlands. *British Journal of Nutrition*. <https://doi.org/10.1017/S0007114511003837>
31. Kiage JN, Sampson UKA, Lipworth L, Fazio S, Mensah GA, Yu Q, Munro H, Akwo EA, Dai Q, Blot WJ, Kabagambe EK (2015) Intake of polyunsaturated fat in relation to mortality among statin users and non-users in the Southern Community Cohort Study. *Nutrition, Metabolism and Cardiovascular Diseases*. <https://doi.org/10.1016/j.numecd.2015.07.006>
32. Marklund M, Wu JHY, Imamura F, Del Gobbo LC, Fretts A, de Goede J, Shi P, Tintle N, Wennberg M, Aslibekyan S, Chen TA, de Oliveira Otto MC, Hirakawa Y, Eriksen HH, Kröger J, Laguzzi F, Lankinen M, Murphy RA, Prem K, Samieri C, Virtanen J, Wood AC, Wong K, Yang WS, Zhou X, Baylin A, Boer JMA, Brouwer IA, Campos H, Chaves PHM, Chien KL, de Faire U, Djoussé L, Eiriksdottir G, El-Abbadi N, Forouhi NG, Michael Gaziano J, Geleijnse JM, Gigante B, Giles G, Guallar E, Gudnason V, Harris T, Harris WS, Helmer C, Hellenius ML, Hodge A, Hu FB, Jacques PF, Jansson JH, Kalsbeek A, Khaw KT, Koh WP, Laakso M, Leander K, Lin HJ, Lind L, Luben R, Luo J, McKnight B, Mursu J, Ninomiya T, Overvad K, Psaty BM, Rimm E, Schulze MB, Siscovick D, Skjelvoll Nielsen M, Smith AV, Steffen BT, Steffen L, Sun Q, Sundström J, Tsai MY, Tunstall-Pedoe H, Uusitupa MJ, van Dam RM, Veenstra J, Monique Verschuren WM, Wareham N, Willett W, Woodward M, Yuan JM, Micha R, Lemaitre RN, Mozaffarian D, Riserus U (2019) Biomarkers of Dietary Omega-6 Fatty Acids and Incident Cardiovascular Disease and Mortality. *Circulation*. <https://doi.org/10.1161/circulationaha.118.038908>
33. Wu JHY, Lemaitre RN, King IB, Song X, Psaty BM, Siscovick DS, Mozaffarian D (2014) Circulating Omega-6 Polyunsaturated Fatty Acids and Total and Cause-Specific Mortality. *Circulation*. <https://doi.org/10.1161/circulationaha.114.011590>
34. Marklund M, Leander K, Vikström M, Laguzzi F, Gigante B, Sjögren P, Cederholm T, de Faire U, Hellenius ML, Riserus U (2015) Polyunsaturated Fat Intake Estimated by Circulating Biomarkers and Risk of Cardiovascular Disease and All-Cause Mortality in a Population-Based Cohort of 60-Year-Old Men and Women. *Circulation*. <https://doi.org/10.1161/circulationaha.115.015607>
35. Harris WS, Tintle NL, Ramachandran VS (2018) Erythrocyte n-6 Fatty Acids and Risk for Cardiovascular Outcomes and Total Mortality in the Framingham Heart Study. *Nutrients*. <https://doi.org/10.3390/nu10122012>
36. Shi F, Chowdhury R, Sofianopoulou E, Koulman A, Sun L, Steur M, Aleksandrova K, Dahm CC, Schulze MB, van der Schouw YT, Agnoli C, Amiano P, Boer JMA, Bork CS, Cabrera-Castro N, Eichmann P, Elbaz A, Farràs M, Heath AK, Kaaks R, Katze V, Keski-Rahkonen P, Masala G, Moreno-Iribas C, Panico S, Papier K, Petrova D, Quirós JR, Ricceri F, Severi G, Tjønneland A, Tong TYN, Tumino R, Wareham NJ, Weiderpass E, Di Angelantonio E, Forouhi NG, Danesh J, Butterworth AS, Kaptoge S (2025) Association of circulating fatty acids with cardiovascular disease risk: analysis of individual-level data in three large prospective cohorts and updated meta-analysis. *European Journal of Preventive Cardiology*. <https://doi.org/10.1093/eurjpc/zwae315>
37. Ren XL, Liu Y, Chu WJ, Li ZW, Zhang SS, Zhou ZL, Tang J, Yang B (2023) Blood levels of omega-6 fatty acids and coronary heart disease: a systematic review and metaanalysis of observational epidemiology. *Critical Reviews in Food Science and Nutrition*. <https://doi.org/10.1080/10408398.2022.2056867>
38. Vene SK, Bork CS, Jakobsen MU, Lundbye-Christensen S, Bach FW, Overvad K, Schmidt EB (2018) Linoleic Acid in Adipose Tissue and Development of Ischemic Stroke: A Danish Case-Cohort Study. *Journal of the American Heart Association*. <https://doi.org/10.1161/jaha.118.009820>
39. Zhang W, Zhou F, Huang H, Mao Y, Ye D (2020) Biomarker of dietary linoleic acid and risk for stroke: A systematic review and meta-analysis. *Nutrition*. <https://doi.org/10.1016/j.nut.2020.110953>
40. Borges MC, Schmidt AF, Jefferis B, Wannamethee SG, Lawlor DA, Kivimäki M, Kumari M, Gaunt TR, Ben-Shlomo Y, Tillin T, Menon U, Providencia R, Dale C, Gentry-Maharaj A, Hughes A, Chaturvedi N, Casas JP, Hingorani AD, the UCLEB Consortium (2020) Circulating Fatty Acids and Risk of Coronary Heart Disease and Stroke: Individual Participant Data Meta-Analysis in Up to 16 126 Participants. *Journal of the American Heart Association*. <https://doi.org/10.1161/jaha.119.013131>
41. Johnson GH, Fritsche K (2012) Effect of Dietary Linoleic Acid on Markers of Inflammation in Healthy Persons: A Systematic Review of Randomized Controlled Trials. *Journal of the Academy of Nutrition and Dietetics*. <https://doi.org/10.1016/j.jand.2012.03.029>
42. Su H, Liu R, Chang M, Huang J, Wang X (2017) Dietary linoleic acid intake and blood inflammatory markers: a systematic review and meta-analysis of randomized controlled trials. *Food & Function*. <https://doi.org/10.1039/c7fo00433h>
43. Ajabnoor SM, Thorpe G, Abdelhamid A, Hooper L (2021) Long-term effects of increasing omega-3, omega-6 and total polyunsaturated fats on inflammatory bowel disease and markers of inflammation: a systematic review and meta-analysis of randomized controlled trials. *European Journal of Nutrition*. <https://doi.org/10.1007/s00394-020-02413-y>
44. Innes JK, Calder PC (2018) Omega-6 fatty acids and inflammation. *Prostaglandins, Leukotrienes and Essential Fatty Acids*. <https://doi.org/10.1016/j.plefa.2018.03.004>
45. Calder PC, Campoy C, Eilander A, Fleith M, Forsyth S, Larsson PO, Schellke B, Lohner S, Szommer A, van de Heijning BJM, Mensink RP (2019) A systematic review of the effects of increasing arachidonic acid intake on PUFA status, metabolism and health-related outcomes in humans. *British Journal of Nutrition*. <https://doi.org/10.1017/S0007114519000692>
46. Wanders AJ, Blom WAM, Zock PL, Geleijnse JM, Brouwer IA, Alsema M (2019) Plant-derived polyunsaturated fatty acids and markers of glucose metabolism and insulin resistance: a meta-analysis of randomized controlled feeding trials. *BMJ Open Diabetes Research & Care*. <https://doi.org/10.1136/bmjdr-2018-000585>
47. Brown TJ, Brainard J, Song F, Wang X, Abdelhamid A, Hooper L (2019) Omega-3, omega-6, and total dietary polyunsaturated fat for prevention and treatment of type 2 diabetes mellitus: systematic review and meta-analysis of randomised controlled trials. *BMJ*. <https://doi.org/10.1136/bmj.l4697>
48. Telle-Hansen VH, Gaundal L, Myhrstad MCW (2019) Polyunsaturated Fatty Acids and Glycemic Control in Type 2 Diabetes. *Nutrients*. <https://doi.org/10.3390/nu11051067>
49. Bjermo H, Iggman D, Kullberg J, Dahlman I, Johansson L, Persson L, Berglund J, Pulkki K, Basu S, Uusitupa M, Rudling M, Arner P, Cederholm T, Ahlström H, Riserus U (2012) Effects of n-6 PUFAs compared with SFAs on liver fat, lipoproteins, and inflammation in abdominal obesity: a randomized controlled trial. *The American Journal of Clinical Nutrition*. <https://doi.org/10.3945/ajcn.111.030114>
50. Rosqvist F, Iggman D, Kullberg J, Cedernaes J, Johansson HE, Larsson A, Johansson L, Ahlström H, Arner P, Dahlman I, Riserus U (2014) Overfeeding Visceral Fat Accumulation and Saturated Fat Causes Distinct Effects on Liver and Visceral Fat Accumulation in Humans. *Diabetes*. <https://doi.org/10.2337/db13-1622>
51. Fridén M, Rosqvist F, Kullberg J, Berglund L, Vessby J, Martinell M, Carlsson PO, Hulthe J, Johansson L, Ahmad N, Johansson HE, Rorsman F, Sundström J, Lind L, Landberg R, Orho-Melander M, Ahlström H, Riserus U (2025) Effects of an anti-lipogenic low-carbohydrate high polyunsaturated fat diet or a healthy Nordic diet versus usual care on liver fat and cardiometabolic disorders in type 2 diabetes or prediabetes: a randomized controlled trial (NAFLDiet). *Nature Communications*. <https://doi.org/10.1038/s41467-025-65613-2>
52. Mousavi SM, Jalilipiran Y, Karimi E, Aune D, Larjani B, Mozaffarian D, Willett WC, Esmailzadeh A (2021) Dietary Intake of Linoleic Acid, Its Concentrations, and the Risk of Type 2 Diabetes: A Systematic Review and Dose-Response Meta-analysis of Prospective Cohort Studies. *Diabetes Care*. <https://doi.org/10.2337/dc21-0438>
53. Seah JYH, Ong CN, Koh WP, Yuan JM, van Dam RM (2019) A Dietary Pattern Derived from Reduced Rank Regression and Fatty Acid Biomarkers Is Associated with Lower Risk of Type 2 Diabetes and Coronary Artery Disease in Chinese Adults. *The Journal of Nutrition*. <https://doi.org/10.1093/jn/nxz164>
54. Jiao J, Liu G, Shin HJ, Hu FB, Rimm EB, Rexrode KM, Manson JE, Zong G, Sun Q (2019) Dietary fats and mortality among patients with type 2 diabetes: analysis in two population based cohort studies. *BMJ*. <https://doi.org/10.1136/bmj.l4009>
55. on behalf of the PUFAs group, Hanson S, Thorpe G, Winstanley L, Abdelhamid AS, Hooper L (2020) Omega-3, omega-6 and total dietary polyunsaturated fat on cancer incidence: systematic review and meta-analysis of randomised trials. *British Journal of Cancer*. <https://doi.org/10.1038/s41416-020-0761-6>
56. Kim Y, Kim J (2020) N-6 Polyunsaturated Fatty Acids and Risk of Cancer: Accumulating Evidence from Prospective Studies. *Nutrients*. <https://doi.org/10.3390/nu12092523>
57. Zhou Y, Wang T, Zhai S, Li W, Meng Q (2016) Linoleic acid and breast cancer risk: a meta-analysis. *Public Health Nutrition*. <https://doi.org/10.1017/s136898001500289x>
58. Yousefi M, Eshaghian N, Heidarzadeh-Esfahani N, Askari G, Rasekhi H, Sadeghi O (2024) Dietary intake and biomarkers of linoleic acid and risk of prostate cancer in men: A systematic review and dose-response meta-analysis of prospective cohort studies. *Critical Reviews in Food Science and Nutrition*. <https://doi.org/10.1080/10408398.2023.2200840>

59. Mourtaz S, Percival BC, Parmar D, Grootveld KL, Jansson P, Grootveld M (2019) Toxic aldehyde generation in and food uptake from culinary oils during frying practices: peroxidative resistance of a monounsaturate-rich algae oil. *Scientific Reports*. <https://doi.org/10.1038/s41598-019-39767-1>
60. Liu ZY, Zhou DY, Li A, Zhao MT, Hu YY, Li DY, Xie HK, Zhao Q, Hu XP, Zhang JH, Shahidi F (2020) Effects of temperature and heating time on the formation of aldehydes during the frying process of clam assessed by an HPLC-MS/MS method. *Food Chemistry*. <https://doi.org/10.1016/j.foodchem.2019.125650>
61. Soundararajan P, Parthasarathy S, Sakthivelu M, Karuppiiah KM, Velusamy P, Gopinath SCB, Raman P (2024) Effects of Consuming Repeatedly Heated Edible Oils on Cardiovascular Diseases: A Narrative Review. *Current Medicinal Chemistry*. <https://doi.org/10.2174/0109298673250752230921090452>
62. Ng CY, Leong XF, Masbah N, Adam SK, Kamisah Y, Jaarin K (2014) Heated vegetable oils and cardiovascular disease risk factors. *Vascular Pharmacology*. <https://doi.org/10.1016/j.vph.2014.02.004>
63. Ke Y, Huang L, Xia J, Xu X, Liu H, Li YR (2016) Comparative study of oxidative stress biomarkers in urine of cooks exposed to three types of cooking-related particles. *Toxicology Letters*. <https://doi.org/10.1016/j.toxlet.2016.05.017>
64. Wallace A, Sutherland W, Mann J, Williams S (2001) The effect of meals rich in thermally stressed olive and safflower oils on postprandial serum paraoxonase activity in patients with diabetes. *European Journal of Clinical Nutrition*. <https://doi.org/10.1038/sj.ejcn.1601250>
65. Zhang Y, Sun Y, Yu Q, Song S, Brenna JT, Shen Y, Ye K (2024) Higher ratio of plasma omega-6/omega-3 fatty acids is associated with greater risk of all-cause, cancer, and cardiovascular mortality: A population-based cohort study in UK Biobank. *eLife*. <https://doi.org/10.7554/elife.90132>
66. Tindall AM, Petersen KS, Skulas-Ray AC, Richter CK, Proctor DN, Kris-Etherton PM (2019) Replacing Saturated Fat With Walnuts or Vegetable Oils Improves Central Blood Pressure and Serum Lipids in Adults at Risk for Cardiovascular Disease: A Randomized Controlled-Feeding Trial. *Journal of the American Heart Association*. <https://doi.org/10.1161/jaha.118.011512>
67. Eichelmann F, Sellem L, Wittenbecher C, Jäger S, Kuxhaus O, Prada M, Cuadrat R, Jackson KG, Lovegrove JA, Schulze MB (2022) Deep Lipidomics in Human Plasma: Cardiometabolic Disease Risk and Effect of Dietary Fat Modulation. *Circulation*. <https://doi.org/10.1161/circulationaha.121.056805>
68. Sellem L, Eichelmann F, Jackson KG, Wittenbecher C, Schulze MB, Lovegrove JA (2023) Replacement of dietary saturated with unsaturated fatty acids is associated with beneficial effects on lipidome metabolites: a secondary analysis of a randomized trial. *The American Journal of Clinical Nutrition*. <https://doi.org/10.1016/j.ajcnut.2023.03.024>
69. Jackson KH, Harris WS, Belury MA, Kris-Etherton PM, Calder PC (2024) Beneficial effects of linoleic acid on cardiometabolic health: an update. *Lipids in Health and Disease*. <https://doi.org/10.1186/s12944-024-02246-2>
70. Lee K, Kurniawan K (2025) Are Seed Oils the Culprit in Cardiometabolic and Chronic Diseases? A Narrative Review. *Nutrition Reviews*. <https://doi.org/10.1093/nutrit/nuae205> ■

70 SOURCES · EVERY DOI RESOLVED · RETRACTION-SCREENED VIA CROSSREF
GROUNDED V3.1.1 · COMPILED 28 AUGUST 2026